

Phases 4 & 5b — Assumptions, Diagnostics & Leakage Repair

WAI Project: Soccer Injury Prediction & Workload Management

Predicting severe injury (28+ days out) in European football, 2020–2025

1. Purpose

This document records the assumptions the modelling rests on, the diagnostic checks run against them, and the decisions taken where an assumption was found to be violated. It is written to be read alongside the notebook (Phase 4 cells), which reproduces every figure quoted here.

The single most consequential finding is leakage. Section 4 documents four features derived from the outcome, three of which were removed from the candidate set. Section 9, added after modelling, documents a second and larger leak that this Phase 4 audit did not catch: five injury-history features built from a career window that overlaps the seasons being predicted. Sections 1–8 are preserved as written at the time; Section 9 records what the audit missed and what correcting it cost.

2. Data assumptions

2.1 Sources and coverage

Dataset	Rows	Role
European Football Injuries 2020–2025	15,603	Primary — one row per injury; defines the target
Transfermarkt injury history	143,195	Player-level prior-injury features
Transfermarkt player profiles	92,671	Height, foot, position; bridges names to player_id
PL injuries & team performance	656	Context only — not merged into the model set

Assumed: the primary dataset is a complete census of recorded injuries across the five leagues and five seasons it covers (20/21 through 24/25). It is not a sample, so no sampling-frame correction is applied. Unrecorded or unreported injuries are invisible to the model — see Section 7.

2.2 Record linkage

The primary and Transfermarkt datasets share no key. Players are matched on lower-cased, whitespace-stripped name after removing the trailing numeric id that Transfermarkt appends to its player_name field (for example "Éverton Silva (100011)" becomes "Éverton Silva").

Link	Result
Primary ↔ Transfermarkt profiles (name)	15,162 of 15,603 matched — 97.2%
Unmatched rows	441 — filled with column medians / "unknown"
Primary ↔ Transfermarkt injury labels	275 of 275 injury types matched — 100%
Injury history aggregated for	34,561 players

Assumed: name identity implies player identity. This will fail for genuine namesakes and for players whose name is transliterated differently between sources. At a 97.2% match rate the residual risk is small but non-zero, and it is not directly measurable without a shared identifier.

3. Target variable

severe_injury = 1 where the injury kept the player out for 28 days or more, else 0. Duration is parsed from the primary dataset's Days field, which is stored as text in the form "43 days".

Property	Value
Definition	days_numeric ≥ 28
Severe (1)	5,546 — 35.5%
Not severe (0)	10,057 — 64.5%
Imbalance ratio	1.8 : 1
Duration range	1 to 1,013 days
Mean / median duration	36.1 / 18.0 days

The 28-day threshold is a modelling choice, not a property of the data. It follows the common clinical convention that separates short-term from long-term absence, and it happens to split the data close to one-third / two-thirds, which is convenient. A different threshold would produce a different target and different results; the choice should be stated as such rather than defended as objective.

Class imbalance is mild. At 35.5% minority the case for synthetic oversampling is weak — SMOTE is designed for minority classes in the 1–10% range and here risks manufacturing noise. Phase 5 will use class_weight="balanced" (and scale_pos_weight ≈ 1.8 for XGBoost) as the primary treatment, with SMOTE tested only as a comparison.

4. Target leakage — features removed

A feature leaks when it is derived from the outcome, or is a consequence of it that would not be known at the moment a prediction is needed. Four features fail this test. Three were in the candidate set and are dropped; days_numeric was never a candidate, and is listed for completeness because it is what the target is computed from.

Feature	Corr. w/ target	Status	Why it leaks
days_numeric	0.585	Never a candidate	Definitional — severe_injury IS (days_numeric ≥ 28)
games_missed	0.574	Dropped	Consequence of the injury, unknown at prediction time
league_risk_rate	0.123	Dropped	Group mean of the target computed over the full dataset, including test rows
position_risk_rate	0.040	Dropped	Group mean of the target computed over the full dataset, including test rows

Corroborating evidence: the correlation between days_numeric and games_missed is 0.935. They are two near-identical restatements of injury duration — that is, of the thing being predicted.

The two risk-rate features are a subtler failure. Encoding a category by its mean outcome is legitimate, but the mean must be computed on the training split alone and then mapped onto the test split. Computed over all rows, as in Phase 2, each test row contributes to the value of its own predictor.

Decision: the three candidate features are dropped from the modelling feature set. If a target-encoded position or league feature is wanted in Phase 5, it must be refitted inside each cross-validation fold. Note that the two risk-

rate features were weak predictors in any case (correlations of 0.123 and 0.040), so little genuine signal is lost — the objection to them is methodological, not empirical.

Consequence to state plainly in the report: with these removed, the strongest single correlation with the target falls from 0.585 to 0.268 (avg_days_missed). That is the honest signal ceiling, and model performance should be judged against it.

5. Statistical assumptions

#	Assumption	Status	Action taken
1	No severe multicollinearity	Partly resolved	age_squared dropped; total_days_missed still VIF 11.07
2	No target leakage	Partly resolved	3 dropped here; 5 more found after modelling (Section 9)
3	Linearity of the logit	Partial	Applies to Logistic Regression only
4	No undue influential outliers	Accepted	Extreme durations retained
5	Independence of observations	Violated	Group-aware CV required
6	Manageable class balance	Mild	class_weight rather than SMOTE

5.1 Multicollinearity

Variance inflation factors were computed across all eighteen candidate features. Three exceed the conventional severe threshold of 10:

Feature	VIF (all 18)	Feature	VIF (all 18)
age_squared	132.99	is_knee	1.22
player_age	121.57	is_muscular	1.17
total_days_missed	11.23	is_illness	1.15
total_injuries	9.77	league_risk_rate	1.10
distinct_injury_types	7.06	season_num	1.09
total_games_missed	6.31	height	1.05
max_days_missed	4.56	position_risk_rate	1.02
is_over_30	3.16		
avg_days_missed	2.65		
injuries_in_dataset	1.65		
games_missed	1.35		

player_age and age_squared are a deterministic pair and inflate each other to absurd levels. age_squared is dropped: it exists to let a linear model capture a curved age effect, but it carries no information player_age does not already hold, and the tree models can represent curvature without it. Removing it returns player_age to a VIF of 2.74.

After dropping age_squared and the three leaking features, the retained fourteen give a maximum VIF of 11.07:

Feature	VIF (retained set)	Reading
total_days_missed	11.07	Still above the severe threshold
total_injuries	9.73	Borderline
distinct_injury_types	7.00	Moderate
total_games_missed	6.00	Moderate
max_days_missed	4.55	Acceptable
player_age	2.74	Acceptable
avg_days_missed	2.54	Acceptable
all remaining nine	< 2.51	Acceptable

This assumption is therefore only partly satisfied. The four Transfermarkt history features are four ways of measuring the same underlying quantity — how much this player has been injured before — so their mutual inflation is expected rather than surprising.

Recommendation: drop `total_days_missed` as well. It is close to the product of `total_injuries` and `avg_days_missed`, both of which are retained, so almost nothing is lost; doing so brings the maximum VIF below 10 and lets the report state the assumption as satisfied rather than as a caveat. This has not been applied to the frozen feature set — it is a judgement call, and the tree models are indifferent to it either way. Logistic Regression is the model that suffers: its coefficients on the four history features should not be interpreted individually while they remain.

5.2 Independence of observations

This assumption is violated by construction. Rows are injuries, not players: 15,603 injuries belong to 4,081 players — 3.82 injuries per player on average, and a single player contributes as many as 21 rows. A conventional random train/test split would place different injuries of the same player on both sides, letting the model recognise the player rather than learn the risk pattern — optimistic scores that would not survive on genuinely unseen players.

Mitigation: Phase 5 must split on player, using `GroupShuffleSplit` or `StratifiedGroupKFold` keyed on `player_name_clean`, so that no player appears in both training and test data.

5.3 Linearity of the logit

Logistic Regression assumes each continuous predictor is linear in the log-odds of the outcome. The Phase 4 notebook cell tests this by binning each continuous feature into deciles, computing the empirical log-odds within each bin, and fitting a straight line. Features with a poor straight-line fit are flagged.

Three of the nine continuous features fail:

Feature	Linearity R^2	Verdict
height	0.284	NON-LINEAR
player_age	0.580	NON-LINEAR
total_games_missed	0.778	NON-LINEAR
injuries_in_dataset	0.865	linear
total_days_missed	0.887	linear
avg_days_missed	0.911	linear
max_days_missed	0.915	linear

Feature	Linearity R^2	Verdict
total_injuries	0.939	linear
distinct_injury_types	0.949	linear

height at $R^2 = 0.284$ is badly non-linear, and player_age at 0.580 is not much better — which is consistent with the Phase 3 finding that severe injuries skew slightly younger rather than rising monotonically with age. Random Forest and XGBoost make no linearity assumption, so this does not disqualify them; it is a concrete prediction that Logistic Regression will under-fit precisely the player-attribute features and should place third on recall. Stating that prediction before Phase 5 and then checking it is a stronger piece of methodology than reporting the ranking after the fact.

5.4 Outliers

435 rows (2.79%) carry at least one value beyond four standard deviations, concentrated in the duration features:

Feature	Rows beyond $ z > 4$	Share
avg_days_missed	145	0.93%
height	131	0.84%
max_days_missed	95	0.61%
total_days_missed	88	0.56%
total_games_missed	69	0.44%

Injury durations extend to 1,013 days against a median of 18. These are extreme but genuine — ruptured cruciate ligaments, complex fractures, long post-operative recoveries. They are not data-entry errors, and at 2.79% of rows they are precisely the cases a severity model exists to identify. They are retained. No winsorising or trimming is applied.

6. Final modelling feature set

Fourteen numeric features survive, alongside six categorical features to be encoded in Phase 5.

Group	Features
Player attributes	player_age, is_over_30, height
Injury characteristics	is_muscular, is_knee, is_illness
Prior history (Transfermarkt)	total_injuries, avg_days_missed, max_days_missed, total_days_missed, total_games_missed, distinct_injury_types
Context	injuries_in_dataset, season_num
Categorical (to encode)	player_position, league, injury_category, season_phase, age_group, foot
Grouping key	player_name_clean (splitting only — not a predictor)

7. Limitations and threats to validity

- Survivorship in the injury record. The dataset contains injuries that were recorded. Minor knocks managed without a reported absence are absent, so the model learns the severity distribution of reported injuries, not of all injuries.

- No exposure data. Minutes played, training load, match congestion and travel are not available. These are among the strongest known predictors of injury in the sports-science literature, and their absence caps what any model built on this data can achieve.
- Prior-injury features are not time-bounded. CONFIRMED AND CORRECTED — see Section 9. The Transfermarkt aggregates summarise a player's whole recorded career, including injuries occurring after the row being predicted. This was recorded here as a mild concern; measurement after modelling showed it was the largest leak in the project, and the features have since been rebuilt as as-of-date windows.
- Name-based linkage. 2.8% of rows did not match and carry imputed medians. Namesakes may be silently conflated.
- Thin categories. Of the 14 position labels, "Midfielder" appears once. Rare categories give unstable per-group estimates and should be collapsed or noted.
- Scope. Five European leagues, five seasons. Generalisation to other leagues, women's football, or youth football is not supported by this data.
- Association, not causation. Every relationship reported is associational. The model can flag which players resemble past severe-injury cases; it cannot establish what causes injury, and squad-rotation decisions taken on its output should be framed accordingly.

8. Implications for Phase 5

Decision	Rationale
Use MODEL_FEATURES (14 numeric + 6 categorical)	Leakage-free; the only defensible basis for reported scores
Split by player, not by row	Observations are not independent (5.2)
class_weight="balanced" / scale_pos_weight ≈ 1.8	Imbalance is mild; SMOTE risks synthetic noise
Judge against a ~ 0.27 correlation ceiling	Sets a realistic expectation for achievable AUC
Report recall on the severe class	A missed severe injury costs a club far more than a false alarm
Refit any target encoding inside CV folds	Prevents reintroducing the Section 4 failure
Consider dropping total_days_missed	Only remaining VIF above 10 (11.07); see 5.1
Expect Logistic Regression to place third	height and player_age are non-linear in the logit (5.3)

Correction to the fourth row, added after Phase 5b: the ~ 0.27 ceiling quoted above was itself leak-inflated. It came from avg_days_missed, whose honest correlation with the target is 0.056. The realistic ceiling was far lower than this table assumed, which is why the corrected AUC of 0.747 is a better result than it first appears.

9. Post-hoc correction — the injury-history leak (Phase 5b)

This section was added after Phase 5. It records a leak that the Phase 4 audit in Section 4 did not detect, why it was missed, how it was measured, and what removing it cost. It is placed last because that is the order in which it was found; it should be read as the most important finding in the document.

9.1 What was missed

The Phase 4 leakage audit screened columns originating in the primary injury dataset. It did not screen the five features that arrived through the Transfermarkt merge: `total_injuries`, `avg_days_missed`, `max_days_missed`, `total_games_missed` and `distinct_injury_types`. Those look like legitimate prior-history features, and were treated as such.

They were built in Phase 2 with a `groupby` over each player's entire Transfermarkt record. That record spans 1973–2025 and therefore contains the 2020–25 injuries the model is asked to classify. Each aggregate averages in the very row being predicted.

9.2 Evidence

Measurement	Value	Interpretation
Transfermarkt injury records	143,173	The population the aggregates are computed over
...falling inside the 2020–25 window	72,810 (50.9%)	Half the "career history" is the modelling window itself
Players whose whole record is inside the window	54%	For most players, history and window are the same thing
Median career injuries per player	3	One row contributes about a third of its own feature value
Rows with no prior injury after correction	12.5%	Genuine first-recorded injuries, correctly given zero

9.3 How much of the signal was the answer

Feature	Corr. (career aggregate)	Corr. (as-of)	Signal lost
<code>avg_days_missed</code>	0.268	0.056	79%
<code>max_days_missed</code>	0.151	0.052	65%
<code>total_games_missed</code>	0.130	0.043	67%
<code>total_injuries</code>	–0.112	–0.034	70%
<code>distinct_injury_types</code>	–0.118	–0.041	65%

Between two-thirds and four-fifths of the apparent predictive signal in every history feature was the target leaking back in.

9.4 The correction

The features were rebuilt with an `as-of` join: for each injury, the aggregates are computed over only those injuries that started strictly earlier. In pandas this is `merge_asof` with `allow_exact_matches=False`, keyed on player id and ordered by injury start date. 97.2% of rows link to a Transfermarkt id, matching the Phase 2 merge rate. All three

models were then refitted on the identical split, grids and random seed, so the difference is attributable to the feature change alone.

9.5 Corrected results

Model	AUC (leaked)	AUC (as-of)	Change	Recall (as-of)
XGBoost	0.8012	0.7472	-0.0539	0.7039
Random Forest	0.7921	0.7435	-0.0486	0.6776
Logistic Regression	0.7758	0.7417	-0.0341	0.6981

Two consequences worth stating in the report. First, the corrected models finish within 0.006 AUC of one another, so the earlier conclusion that XGBoost was the better model does not survive: tree ensembles exploit leaked features more effectively than a linear model, and most of their apparent advantage was that exploitation. The Section 8 expectation that Logistic Regression would place third is technically borne out but by a margin too small to claim as a confirmed prediction.

Second, the drivers changed entirely. Before correction the model was led by `avg_days_missed` (mean `|SHAP|` 0.69). After correction it is led by injury type — `is_illness` at 0.48, `injury_category_other` at 0.33, `is_knee` at 0.14 — with the strongest history feature down at 0.10. The defensible finding is that what kind of injury a player sustains predicts severity, and who the player is barely does. This is consistent with the Phase 3 chi-square results, where injury category scored 2165.6 against position's 24.7.

9.6 What this means for the method

- A leakage audit must cover every column in the modelling frame, not only those from the primary source. Features acquired through a merge inherit the merge's time window, and that window is rarely the one the audit assumes.
- Any feature summarising a subject's history must be defined as-of the prediction moment. "The player's injury record" is not a well-defined quantity until a cut-off date is attached to it.
- A feature that dominates a SHAP ranking deserves suspicion before celebration. The size of `avg_days_missed`'s lead was the clue that led to this section.
- Two items remain open: `injuries_in_dataset` still counts a player's injuries across the whole window and should be made as-of or dropped, and the decision-threshold sweep was run against the leaked model and needs repeating.

Every figure in this document is reproduced by the Phase 4 and Phase 5b cells in `WAI_Soccer_Injury_Prediction.ipynb`. Sections 1–8 record the state of the analysis before modelling; Section 9 records the correction made after it.